

Reconstructing permutations of order eleven from five-element patterns

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Abstract

The k -deck of a permutation is the multiset of order patterns of all its k -element subsequences. We give a computer-assisted proof that the least deck size determining every permutation of order eleven is $s(11) = 5$. More precisely, sixteen position–rank marginals of the five-deck already distinguish all $11! = 39,916,800$ permutations. An exhaustive exact-integer computation and a separate implementation both find that many distinct marginal vectors. We describe the counting identity, encoding and coverage checks underlying this finite step, and give an explicit four-deck collision for the lower bound. The result also implies injectivity of the six- and seven-deck maps at order eleven.

1 Statement and context

Write $[n] = \{1, \dots, n\}$ and let S_n be its permutations in one-line notation. For a sequence a of distinct numbers, its standardization $\text{st}(a)$ replaces each entry by its rank among the entries of a . For $x \in S_n$, $1 \leq k \leq n$ and $\pi \in S_k$, set

$$c_\pi(x) = \#\{i_1 < \dots < i_k : \text{st}(x_{i_1}, \dots, x_{i_k}) = \pi\}.$$

The multiset $D_k(x)$ has multiplicity $c_\pi(x)$ at π ; in particular, it retains repeated patterns. Define

$$s(n) = \min\{k \in [n] : D_k(x) = D_k(y) \implies x = y \text{ for all } x, y \in S_n\}.$$

For $x \in S_{11}$ define the five-deck marginals

$$M_{jr}(x) = \sum_{\substack{\pi \in S_5 \\ \pi_j = r}} c_\pi(x), \quad 1 \leq j, r \leq 5. \quad (1)$$

Thus M_{jr} counts five-subsequences whose j th entry has rank r .

Theorem 1 (Computer-assisted). *The map $x \mapsto (M_{jr}(x))_{1 \leq j, r \leq 4}$ is injective on S_{11} . Consequently $s(11) = 5$: the five-, six- and seven-deck maps on S_{11} are injective, whereas the four-deck map is not.*

Ma, Zhong and Zhang [1] give $5 \leq s(11) \leq 8$ in their published Table 2. Their preprint, Section IV-A, identifies $s(11)$ as the first undetermined value in its table, and Lemma 4.3 establishes $s(9), s(10) \leq 5$. Their Lemma 4.1 and Corollary 4.1 use position marginals for the smallest and largest selected ranks. The present result determines the value at order eleven by checking the sixteen marginals in Theorem 1. The general use of deck marginals and the lower-bound construction

are inherited ideas. Exact finite certification in permutation enumeration also appears in Zeng [2]; that work supplies adjacent computational context concerning cycle lengths, and no reconstruction result from it is used here. Our comparison is with the inspected sources; it does not assert global priority or any result for orders greater than eleven.

2 Counting the marginals

For $v \in [11]$, let $B_v(x)$ be the set of values preceding v in x . For $B \subseteq [11] \setminus \{v\}$ define

$$T_{jr}(B, v) = \# \left\{ A \subseteq [11] \setminus \{v\} : \begin{array}{l} |A| = 4, \quad |A \cap B| = j - 1, \\ |A \cap [v - 1]| = r - 1 \end{array} \right\}. \quad (2)$$

Here $[0] = \emptyset$.

Lemma 2 (Anchor identity). *For all $x \in S_{11}$ and $1 \leq j, r \leq 5$,*

$$M_{jr}(x) = \sum_{v=1}^{11} T_{jr}(B_v(x), v). \quad (3)$$

Proof. The value set $A \cup \{v\}$ determines exactly one subsequence of x . Within it, v has position $1 + |A \cap B_v(x)|$ and rank $1 + |A \cap [v - 1]|$. Hence its contribution is counted by (2). For fixed j , every five-subsequence has exactly one anchor in position j . Summing over the anchor value proves (3) without imposing any further order condition on A . $\square$

Every marginal is between 0 and $\binom{11}{5} = 462$. Each row and each column of the 5×5 marginal matrix sums to 462, because every five-subsequence has one entry in each position and one entry of each rank. Thus the first four rows and columns determine the remaining nine entries by subtraction. Only the simpler implication

$$D_5(x) = D_5(y) \implies (M_{jr}(x))_{j,r \leq 4} = (M_{jr}(y))_{j,r \leq 4} \quad (4)$$

is needed to pass from marginal injectivity to deck injectivity.

3 The exhaustive finite step

Proposition 3 (Exact computation). *Among the 39,916,800 permutations of $[11]$, the sixteen-vectors $(M_{jr})_{1 \leq j, r \leq 4}$ take 39,916,800 distinct values.*

We give the algorithm and its arithmetic justification, followed by the recorded output on which Proposition 3 depends. The full source and replay driver are supplied with the article.

Transitions and exact keys. For each of the $11 \cdot 2^{10} = 11,264$ possible pairs (B, v) , the program enumerates all $\binom{10}{4} = 210$ four-element value sets avoiding v . It calculates the two intersection sizes in (2), keeping $1 \leq j, r \leq 4$. During a left-to-right scan of a permutation, the set of entries already seen is exactly B_v . The sum of the eleven transition vectors therefore equals the required marginal vector by Lemma 2.

Index its coordinates as $m_{4(j-1)+(r-1)} = M_{jr}$ for $j, r \leq 4$. The program represents them by three unsigned 64-bit words:

$$K_w = \sum_{\substack{0 \leq i \leq 15 \\ \lfloor i/6 \rfloor = w}} m_i 512^{i \bmod 6}, \quad w = 0, 1, 2. \quad (5)$$

There are six, six and four digits in the three words. Every digit is less than 512, so the encoding is injective. A word uses at most 54 bits. Encoded transitions can be added word by word: contributions are nonnegative, and each partial coordinate sum is at most its final value, which is at most 462. There are no interdigit carries or word overflows. Equality compares all three words; it uses no fingerprints.

Domain and collision checks. Starting at the increasing permutation, the program repeatedly calls `std::next_permutation` until exhaustion. It checks that the four-bit-per-entry encodings of the input permutations strictly increase, that there are exactly $11!$ inputs, and that each possible first entry occurs $10! = 3,628,800$ times. Distinct inputs belonging to S_{11} with total count $11!$ constitute the entire domain. No symmetry quotient is used. The algorithm can be summarized as follows:

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Precompute every transition  $T(B, v)$  using the four-element value sets.
For every permutation  $x$  of  $[11]$ :
    set  $B = \emptyset$  and  $K = (0, 0, 0)$ ;
    for each successive entry  $v$  of  $x$ , add the encoded  $T(B, v)$  to  $K$  and set  $B \leftarrow B \cup \{v\}$ ;
    append  $K$  to the list of keys.
Sort the keys and count equal adjacent pairs.

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Equal keys must become adjacent after sorting. The completed computation reports 39,916,800 keys and zero equal adjacent pairs. In view of (3) and the injective encoding, this is the finite verification of Proposition 3.

Separate exact implementation. A second exact verifier computes each transition using the coefficients of

$$\prod_{w \in [11] \setminus \{v\}} (1 + z a^{1_{w \in B}} b^{1_{w < v}}). \quad (6)$$

The coefficient of $z^4 a^{j-1} b^{r-1}$ is $T_{jr}(B, v)$: choosing the nonconstant term in four factors selects precisely the set A . A dynamic program multiplying these factors in descending z -degree computes the coefficients with integer additions. The sum of the 25 degree-four coefficients is checked to be 210 for each transition.

This implementation stores sixteen raw unsigned 16-bit coordinates. Even the loose accumulation bound $11 \cdot 210 = 2310$ is below 2^{16} , and every completed coordinate is additionally checked to be at most 462. It enumerates inputs by Heap’s swap algorithm, validates each permutation, and marks its Lehmer rank in a bitset that rejects repeated ranks. The count of marked ranks is $11!$, independently certifying complete coverage. It then sorts the raw vectors and checks their order and adjacent equality. Its results agree with the packed-key computation:

Quantity	Packed keys	Raw coordinates
Permutations covered	39,916,800	39,916,800
Distinct marginal vectors	39,916,800	39,916,800
Equal adjacent pairs	0	0
Direct-oracle comparisons	512	4,875

For the oracle comparisons, each marginal is computed directly from the 462 five-position subsets. The raw-coordinate verifier uses all inputs whose Lehmer rank is divisible by 8191, together with the last rank; the packed-key verifier uses 512 deterministic cases, including the increasing and decreasing permutations. These checks support implementation correctness; the full enumeration

supplies the universal finite step. The raw-coordinate run also completed with undefined-behavior sanitization enabled and no sanitizer diagnostic.

4 The lower bound and completion of the proof

Consider the distinct permutations

$$\begin{aligned}x &= (6, 8, 5, 7, 3, 1, 4, 2, 9, 10, 11), \\y &= (7, 5, 8, 6, 2, 4, 1, 3, 9, 10, 11).\end{aligned}$$

They extend the four-deck collision 68573142, 75862413 recorded in [1, Section IV-A of the preprint] by appending three increasing maxima. Direct enumeration of the 330 four-position subsets gives the common multiplicities below. A pattern such as 1234 denotes $(1, 2, 3, 4)$.

Pattern	Count	Pattern	Count	Pattern	Count
1234	26	2314	42	3412	9
1243	0	2341	0	3421	13
1324	6	2413	1	4123	0
1342	0	2431	4	4132	4
1423	0	3124	42	4213	4
1432	0	3142	1	4231	8
2134	72	3214	72	4312	13
2143	0	3241	4	4321	9

The counts sum to 330, and the supplement verifies both decks individually using exact integer arithmetic. This is an explicit known lower-bound construction, included to make the conclusion self-contained.

Lemma 4 (Subdeck identity). *For $1 \leq \ell \leq k \leq n$, $\sigma \in S_\ell$ and $x \in S_n$,*

$$\sum_{\pi \in S_k} c_\pi(x) c_\sigma(\pi) = \binom{n-\ell}{k-\ell} c_\sigma(x). \quad (7)$$

Consequently equality of k -decks implies equality of ℓ -decks.

Proof. Count pairs of position subsets $I \subseteq J \subseteq [n]$ with $|I| = \ell$, $|J| = k$ and pattern σ on I . First choosing J gives the left side, since standardization preserves all relative comparisons. First choosing I gives the right side: each I is contained in exactly $\binom{n-\ell}{k-\ell}$ possible J . The binomial factor is positive, so it can be divided out of the multiplicity identity. $\square$

Proof of Theorem 1. Proposition 3 proves the stated sixteen-marginal injectivity. Implication (4) gives five-deck injectivity. The displayed four-deck collision and Lemma 4 rule out every $k \leq 4$, hence $s(11) = 5$. Finally, applying the lemma with $n = 11$, $\ell = 5$ and $k = 6, 7$ (with binomial factors 6 and 15) proves six- and seven-deck injectivity as well. $\square$

Reproducibility and scope of the computation

The supplementary directory contains the unmodified packed-key C++ source, the separate raw-coordinate C++ source, a Python lower-bound checker, the recorded exact outputs, and a replay

driver. Running `python3 replay.py` from that directory compiles both C++17 programs, runs them sequentially in a separate output directory, and checks the decisive integer counts against the supplied records. The recorded computations used Apple Clang 16 on an ARM64 macOS system. The packed and raw key arrays occupy approximately 0.96 and 1.28 GB, respectively, before auxiliary storage. The driver requires Python 3, a compatible C++17 compiler and sufficient memory; no network access or external data are needed.

The proof is computer-assisted: the finite injectivity assertion depends on the source programs, compiler, exact integer execution, sorting and hardware. The supplied implementations differ in transition calculation, enumeration and key representation, but were run on the same machine and compiler family. They are not proof-assistant formalizations. No claim for $n > 11$ follows from the computation. Neither the published upper bound 8 nor an assumed correctness of earlier published enumerations is needed for the proof.

AI assistance. OpenAI Codex assisted with the mathematical reduction, verification code, manuscript preparation and checks. A separate model context inspected and replayed the argument and produced the raw-coordinate verifier. This is a separate computational review pass, not journal peer review or formal verification. The exact procedures and their scope are specified above and in the supplementary files.

References

- [1] Yiming Ma, Wenjie Zhong and Xiande Zhang, *Improvements on permutation reconstruction from minors*, Advances in Mathematics of Communications **21** (2026), 95–106. doi:10.3934/amc.2025044. Preprint version: arXiv:2411.12718v1 (2024).
- [2] Zijian Zeng, *The Distinct-Cycle-Length Probability is Strictly Decreasing after 30*, SSRN working paper (2026). doi:10.2139/ssrn.7380519.